Free Will as Computational Necessity: A Language-Theoretic and Concurrency-Control Proof of Non-Deterministic Agency

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Abstract

We present three independent formal proofs that non-deterministic agency ("free will") is not a metaphysical luxury but a *computational necessity* for any non-trivial dynamical system. First, using Language-Theoretic Security (LangSec), we model free will as a *parser*—a non-deterministic finite automaton that maps environmental inputs to behavioral outputs through an internal state that evolves over time. We prove that deterministic parsing ($W = 0$) produces zero novelty: the system converges to a static fixed point with no new information generation. Second, using thermodynamic entropy analysis, we prove that without non-deterministic input injection, any closed system monotonically approaches heat death ($\Delta S \to \max$), and that free will is the *only mechanism* capable of locally decreasing entropy ($\Delta S < 0$). Third, using Optimistic Concurrency Control (OCC) from distributed systems theory, we resolve the classical sovereignty-vs-agency paradox: we demonstrate that global determinism (omniscience) and local non-determinism (free will) are not merely compatible but are the defining architectural features of any non-locking, eventually consistent distributed system.

Keywords: free will, determinism, compatibilism, language-theoretic security, optimistic concurrency control, entropy, non-deterministic automata, distributed systems.

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1 Introduction

These foundational principles operate on established invariants Penrose [1989], Shannon [1948b].

The free will debate is among the oldest in philosophy, typically framed as a binary opposition between libertarian free will (agents can genuinely choose) and hard determinism (all events are causally necessitated by prior states). Compatibilism attempts a middle path, but its formulations are often criticized as definitional maneuvers rather than formal proofs [Kane, 2005].

We approach the problem from a fundamentally different direction: we do not ask *whether* free will exists, but rather *whether the system can function without it.* Our answer is negative, and we prove it through three independent formal frameworks:

[label=(i)]

1. **Language-Theoretic Security (LangSec).** We model agents as parsers of environmental data. A deterministic parser produces identical outputs for identical inputs—no novelty, no meaning. A non-deterministic parser (free will) is required for the system to generate new information [Bratus et al., 2014].

2. **Thermodynamic Entropy.** The Second Law of Thermodynamics guarantees that closed systems converge to maximum entropy. Without an external entropy-reducing mechanism, the system dies. We prove that free will—the non-deterministic choice to create order—is the only endogenous mechanism available [Shannon, 1948a, Clausius, 1865].

3. **Optimistic Concurrency Control (OCC).** In distributed databases, OCC allows multiple transactions to execute simultaneously without locks. Each transaction acts *optimistically* (freely), and a global coordinator resolves conflicts at commit time. We prove that this architecture is isomorphic to the sovereignty-agency relationship: global determinism (the coordinator) and local non-determinism (the transactions) coexist by design [Kung and Robinson, 1981].

1.1 Main Results

Theorem 1.1 (The Novelty Requirement)**.** *Let be a dynamical system with decision function $D : \times\Sigma^* \to \Gamma^*$. If D is deterministic (i.e., a function of inputs alone with no internal non-deterministic state), then the information-theoretic novelty of is zero:* $I_{novel}() = 0$.

Theorem 1.2 (The Entropy Injection Theorem)**.** *In any closed system governed by the Second Law of Thermodynamics, $\Delta S \geq 0$. Local entropy reduction ($\Delta S_{local} < 0$) requires an endogenous non-deterministic mechanism—a "choice" that exports entropy to the environment. Without such a mechanism, converges monotonically to heat death.*

Theorem 1.3 (The OCC Compatibility Theorem)**.** *In an Optimistic Concurrency Control architecture, global consistency (all transactions commit to a valid final state) and local non-determinism (each transaction executes without synchronous locks) are simultaneously achievable. The temporal gap $\Delta t = t_{commit} - t_{execute}$ between local action and global integration does not compromise either property.*

2 The LangSec Proof: Free Will as a Parser

2.1 Agents as Automata

In Language-Theoretic Security [Bratus et al., 2014], the security of a system depends on whether its input-processing components (parsers) correctly recognize the formal language of valid inputs. We extend this framework to model cognitive agents.

Definition 2.1 (Decision Function)**.** An agent's *decision function* is a mapping $D : \times\Sigma^* \to \Gamma^*$, where is the set of system constraints (laws of physics, social norms), Σ^* is the set of all possible input strings (events, stimuli), and Γ^* is the set of all possible output strings (actions, responses). The "will" W of an agent is defined as the cardinality of valid parse paths:

$$W = \int |D(x)| \, dx, \tag{1}$$

where $|D(x)|$ is the number of distinct valid outputs for input x within the constraint set .

Proof of thm:novelty. If D is deterministic, then $|D(x)| = 1$ for all $x \in \Sigma^*$. Therefore $W = \int 1 \, dx = ||$, which measures the size of the constraint set but produces *zero choice-entropy*:

$$H_{\text{choice}} = -\sum_i p_i \log p_i = -1 \cdot \log 1 = 0. \tag{2}$$

Since $H_{\text{choice}} = 0$, the system generates no new information beyond what is determined by its initial conditions and constraints. $I_{\text{novel}}() = 0$. □

2.2 The Parser Interpretation

Under this model, an agent's free will is literally its parser. The environment sends a string of events (data), and the agent *chooses* how to interpret (parse) that data. Parsing through the lens of "peace" generates one output; parsing through "fear" generates another. The constraints are fixed (physical laws, logical boundaries), but the parse *within* those constraints is non-deterministic.

[Connection to Tzimtzum] In Kabbalistic theology, the concept of *Tzimtzum* (divine self-contraction) posits that the Creator "withdrew" absolute control to create a "void" in which free agents could exist [Scholem, 1941]. In our framework, is the boundary of this void: the constraints are fixed, but within them, $W > 0$—the agent has genuine agency.

3 The Thermodynamic Proof: Entropy and Agency

Proof of thm:entropy. By the Second Law [Clausius, 1865], for any closed system: $\Delta S_{\text{total}} \geq 0$. However, open subsystems within the closed universe can achieve $\Delta S_{\text{local}} < 0$ by exporting entropy to their environment, provided the total entropy still increases.

The question is: what *mechanism* drives local entropy reduction? In biology, this mechanism is *choice*—the non-deterministic selection of order-creating actions from a set of alternatives. A cell that "chooses" to replicate its DNA rather than degrade reduces local entropy. An agent that "chooses" to build rather than destroy reduces local entropy.

Without $W > 0$ (non-deterministic agency), the system has no endogenous mechanism for selecting entropy-reducing trajectories. It can only follow the gradient of maximum entropy: heat death.

Therefore: $W > 0$ (free will) is the necessary condition for $\Delta S_{\text{local}} < 0$ (local order creation). □

Definition 3.1 (Moral Vector)**.** The *moral vector* $\vec{V}$ of an action is defined as the ratio of information gain to entropy change:

$$\vec{V} = \frac{\Delta I}{\Delta S}, \tag{3}$$

where ΔI is the change in mutual information between the agent and its environment, and ΔS is the change in system entropy. If $\vec{V} > 0$, the action is *constructive* (increases information density while reducing entropy). If $\vec{V} < 0$, it is *destructive*.

4 The OCC Proof: Sovereignty and Agency

4.1 The Sovereignty–Agency Paradox

The classical theological formulation: if an omniscient coordinator ("God") knows all future states, do local agents have genuine choice? Two failure modes arise:

[label=(a)]

1. **Fatalism (Dead-Code Exception):** All actions are pre-compiled; local processing is meaningless.
2. **Anxiety (Out-of-Memory Panic):** The local agent believes it alone maintains global coherence and must continuously self-validate.

Both modes are architectural errors. We resolve them via OCC.

4.2 The OCC Architecture

Definition 4.1 (OCC Transaction)**.** In Optimistic Concurrency Control [Kung and Robinson, 1981], multiple transactions $\{T_1, \ldots, T_n\}$ execute simultaneously on a shared database without acquiring synchronous locks. Each transaction T_i:

[label=(i)]

1. Reads a snapshot σ_i of the global state;
2. Executes local mutations M_i based on σ_i;
3. Submits (M_i, σ_i) for validation at commit time.

The global coordinator validates all transactions by checking for conflicts (two transactions modifying the same state). Conflicting transactions are rolled back and retried; non-conflicting transactions are committed atomically.

Proof of thm:occ. We establish the isomorphism:

OCC Component	Sovereignty–Agency Mapping
Global Database	The state of reality
Transaction T_i	An individual agent's life-trajectory
Local Mutation M_i	A free choice / action
Snapshot σ_i	The agent's perception of state
Commit Validation	Divine judgment / conflict resolution
Holy Latency Δt	The gap between action and consequence

In OCC, the following properties hold simultaneously:

[label=(ii)]

1. **Local Non-Determinism:** Each T_i executes without locks—it genuinely "chooses" its mutations M_i based on its local state, without being constrained by concurrent transactions.
2. **Global Consistency:** The coordinator guarantees that the final committed state is serializable—equivalent to some sequential execution order. No inconsistency survives the commit phase.
3. **Eventual Convergence:** The system converges to a consistent state even if individual transactions fail and retry.

These are precisely the properties required to resolve the sovereignty–agency paradox: the coordinator (sovereignty) and the transactions (agency) coexist by architectural design. Neither property compromises the other.

The "Holy Latency" $\Delta t = t_{\text{commit}} - t_{\text{execute}}$ explains why free actions do not receive immediate environmental feedback: the global state will not reflect the local commit until all cross-agent conflicts are resolved. This latency is not absence of coordination; it is the computational cost of eventual consistency in a massively parallel system. □

5 Discussion

5.1 The Unity of the Three Proofs

The three proofs are independent but convergent:

- **LangSec** proves that free will is necessary for *informational novelty.*
- **Thermodynamics** proves that free will is necessary for *order creation.*
- **OCC** proves that free will is *compatible* with global determinism.

Together, they establish that non-deterministic agency is simultaneously *required* (by information theory and thermodynamics) and *compatible* (with omniscient coordination).

5.2 Limitations

1. The LangSec model treats agents as parsers, which is a simplification. Real cognitive systems are more complex than finite-state automata.
2. The thermodynamic proof assumes that entropy reduction requires "choice"—but biological entropy reduction is sometimes driven by physical gradients (e.g., crystallization) without apparent agency.
3. The OCC proof establishes compatibility but does not prove existence. It shows that sovereignty and agency *can* coexist, not that they *do* in our universe.

6 Conclusion

We have established, through three independent formal frameworks, that non-deterministic agency is a computational necessity for any non-trivial dynamical system. Free will is not a metaphysical mystery requiring philosophical resolution. It is a *formal requirement* for:

1. information-theoretic novelty (zero novelty without it),
2. thermodynamic sustainability (heat death without it), and
3. architectural consistency in distributed systems (achievable with it).

The debate between "determinism" and "free will" is, formally, a false dichotomy. Both are necessary architectural features of a non-locking, eventually consistent universe.

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